Equity implications of federal–local cost-sharing in flood buyouts

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Abstract

Climate-driven flood risk increasingly necessitates managed retreat through government buyout programmes, yet empirical evidence documents substantial racial and economic disparities in programme implementation. Here we develop a three-level Stackelberg game to analyse how federal–local cost-sharing arrangements generate inequitable outcomes through strategic interactions among federal authorities, local governments, and heterogeneous homeowners. Our model reveals three distinct mechanisms driving inequity: differential discount rates across income groups, local governments' tax-base preservation incentives, and participation thresholds that exclude fiscally constrained communities. Numerical analysis of 34,493 households across nine flood-prone US regions demonstrates that the current Federal Emergency Management Agency 75/25 cost-sharing arrangement produces a relocation ratio gap of 0.26—low-income households relocate at roughly one-quarter the rate of high-income households. Achieving near-equity requires federal cost shares of at least 85%, though equity-weighted mechanisms can attain similar outcomes at 25% lower cost. These findings provide a theoretical foundation for understanding observed disparities and identify policy levers for more equitable climate adaptation.

Keywords: flood buyouts, managed retreat, Stackelberg game, environmental justice, climate adaptation

Introduction

Managed retreat through property buyouts has emerged as a critical climate adaptation strategy as flood risk intensifies across the United States Siders et al. [2019], Mach et al. [2019]. The Federal Emergency Management Agency (FEMA) administers primary buyout programmes through the Hazard Mitigation Grant Program (HMGP) and Building Resilient Infrastructure and Communities (BRIC) initiative, which typically operate on a 75/25 federal–local cost-sharing basis. However, mounting empirical evidence reveals troubling disparities: Elliott et al. Elliott et al. [2021] documented that buyout funding disproportionately flows to wealthier, white communities, while Greer and Binder Greer and Binder [2017] found systematic barriers facing low-income participants.

These disparities likely emerge from incentive structures embedded within cost-sharing arrangements, yet existing theoretical frameworks inadequately capture the multi-level strategic dynamics at play. The foundational game-theoretic model of Bier et al. Bier et al. [2019] treats government as a unified decision-maker, abstracting away from the divergent objectives between federal and local authorities that may drive

observed inequities. Local governments facing a 25% cost share confront competing pressures: reducing disaster exposure through buyouts versus preserving property tax revenue and maintaining community stability.

We address this gap by developing a three-level stochastic Stackelberg game that explicitly models strategic interactions among federal government, local governments, and heterogeneous homeowners under climate uncertainty. The federal government acts as Stackelberg leader, setting cost-sharing parameters that shape local government participation decisions and, ultimately, which households receive buyout offers. Our framework identifies three distinct channels through which current policy generates inequitable outcomes and evaluates alternative mechanisms for improving distributional outcomes.

Results

Theoretical framework and equilibrium characterization

Our model captures a sequential game played under climate uncertainty. The federal government first announces a cost-sharing ratio $\alpha \in [0,1]$ specifying its share of buyout costs and a maximum subsidy cap $\bar{S}$. Local governments then simultaneously decide whether to participate and what subsidy levels to offer. Finally, homeowners choose optimal relocation timing given offered subsidies.

Homeowners are characterized by type $\theta_h = (V_h, r_h, \ell_h)$ representing house value, discount rate, and location. Following Warner and Pleeter Warner and Pleeter [2001] and Bier et al. Bier et al. [2019], we model low-income households as having higher effective discount rates ($r^{\text{high}} = 18\%$) than high-income households ($r^{\text{low}} = 12\%$) due to credit constraints. This assumption, while necessarily simplifying the heterogeneous financial circumstances of low-income households, captures the well-documented empirical regularity that credit-constrained individuals exhibit higher implicit discount rates in revealed-preference studies Hausman [1979], Frederick et al. [2002]. The magnitude of this differential proves central to understanding inequitable outcomes, and we examine sensitivity to alternative specifications in our robustness analysis.

The homeowner's optimal relocation timing balances discounted relocation costs against cumulative expected flood damages. For household h receiving subsidy S, the optimal relocation year satisfies a threshold condition where the present value of future flood damages equals net relocation costs (see Methods). Higher subsidies induce earlier relocation, but the magnitude of this effect depends critically on discount rates: a uniform subsidy shifts low-discount-rate households' decisions more than high-discount-rate households'.

Local governments minimize expected costs comprising their buyout cost share, administrative expenses, disaster-related expenditures, and a tax-base preservation term weighted by parameter λ_j. This specification captures one dimension of political economy—fiscal pressures to maintain property tax revenue—while abstracting from other relevant factors including electoral cycles, constituent preferences, and administrative capacity constraints. The first-order condition for optimal subsidy S_j^* reveals that higher federal cost shares α induce higher local subsidy offers, while stronger tax-base concerns reduce subsidies overall (Supplementary Information, Theorem S2).

We characterize subgame perfect Nash equilibrium through backward induction. Equilibrium existence follows from standard continuity arguments (Supplementary Information, Theorem S3). The federal govern-

ment's optimal policy trades off aggregate social costs against equity constraints, with the equity constraint binding for sufficiently stringent equity targets.

Mechanisms generating inequity

Our theoretical analysis identifies three distinct channels through which current cost-sharing arrangements generate inequitable outcomes (Fig. 5).

The *discount rate channel* operates through differential subsidy effectiveness across income groups. Because low-income households discount future flood damages more heavily, they require larger subsidies to induce equivalent relocation timing. Specifically, to induce relocation in year k for a household with discount rate r, the required subsidy equals relocation costs minus discounted future damages—a quantity that increases with r. Under uniform subsidies, low-income households therefore relocate later or not at all.

The *tax-base preservation channel* arises from local governments' fiscal incentives. When $\lambda_j > 0$, local governments prefer buyouts that minimise tax-base erosion. This creates complex targeting dynamics: local officials may favour lower-value properties to preserve aggregate tax revenue, but low-income households do not necessarily occupy lower-value properties, particularly in gentrifying neighbourhoods or when lower-income families reside in inherited homes.

The *participation channel* reflects variation in local governments' ability to meet cost-sharing requirements. The critical threshold α_j^{crit} above which jurisdiction j participates depends inversely on fiscal capacity and directly on administrative costs. Communities with limited fiscal resources—often lower-income communities—require higher federal cost shares to participate. Below their threshold, these communities receive no buyout programme at all.

Numerical analysis

We apply our framework to a multi-region simulation encompassing 34,493 households across nine flood-prone US regions: four New York City areas (Brooklyn, Staten Island, Queens, Lower Manhattan), Houston, New Orleans, Miami-Dade, Charleston, and Norfolk. Property values range from $245,000 (Norfolk) to $1.85 million (Lower Manhattan), low-income population shares range from 15% (Lower Manhattan) to 62% (New Orleans), and 100-year floodplain exposure ranges from 35% (Brooklyn) to 75% (New Orleans). This heterogeneity allows us to examine how regional characteristics interact with federal policy to shape distributional outcomes.

Figure 1 presents our central finding: the relationship between federal cost-share ratio and the relocation ratio gap (RRG), defined as the ratio of low-income to high-income relocation rates. Under the current FEMA 75/25 arrangement ($\alpha = 0.75$), we observe RRG = 0.26, indicating that low-income households relocate at roughly one-quarter the rate of high-income households. At this policy, only 191 of 17,123 low-income households (1.1%) participate in buyouts, compared with 693 of 17,370 high-income households (4.0%).

Achieving near-equity (RRG ≥ 0.70) requires federal cost shares of at least 85%, which increases federal expenditure from $82 million to $375 million—a 4.6-fold increase. Beyond $\alpha = 0.90$, diminishing returns

set in: RRG increases from 0.87 to 0.93 as α rises from 0.90 to 1.00, while costs increase by an additional \$176 million. The transition from inequitable to near-equitable outcomes thus occurs over a relatively narrow policy range.

Regional heterogeneity substantially shapes these aggregate patterns (Fig. 3). Lower Manhattan achieves high participation rates at lower α values due to high property values and severe flood exposure, while lower-income regions like Norfolk and New Orleans require $\alpha \geq 0.85$ for meaningful participation. This variation reflects both the tax-base channel (Lower Manhattan's high property values create strong preservation incentives) and the participation channel (Norfolk's limited fiscal capacity raises its participation threshold).

Sensitivity analysis

We assess robustness to key parameter assumptions through systematic sensitivity analyses (Extended Data Fig. 1). The discount rate differential between income groups proves quantitatively important: under low heterogeneity ($r^{\text{high}} = 14\%$ versus $r^{\text{low}} = 12\%$), achieving RRG ≥ 0.70 requires only $\alpha = 0.78$, while extreme credit constraints ($r^{\text{high}} = 25\%$ versus $r^{\text{low}} = 8\%$) require $\alpha \geq 0.92$. This sensitivity underscores the importance of accurately characterising household discount rates in policy design.

Subsidy cap variations interact with cost-sharing ratios. At the baseline cap of \$250,000, increasing α from 0.75 to 0.90 improves RRG from 0.26 to 0.87. At a higher cap of \$400,000, the same α increase yields RRG improvements from 0.31 to 0.91, suggesting complementarity between the two policy instruments (Extended Data Fig. 2).

Climate scenario uncertainty does not qualitatively alter our conclusions. Under pessimistic scenarios (1.0m sea-level rise by 2100), higher flood damages make buyouts more attractive for all households, but relative inequity persists. The RRG under current policy ranges from 0.24 (optimistic scenario) to 0.29 (pessimistic scenario), a modest variation that does not change the fundamental policy implications.

Alternative policy mechanisms

We evaluate three alternative policy designs against the current uniform cost-sharing arrangement (Fig. 4).

Equity-weighted cost sharing sets jurisdiction-specific federal shares according to $\alpha_j = \bar{\alpha} + \gamma(MHI_{\text{nat}} - MHI_j)$, where MHI_j denotes median household income in jurisdiction j and $\gamma > 0$ is a progressivity parameter. This mechanism directs higher federal support to lower-income communities, addressing the participation channel directly. With $\gamma = 0.10$, equity-weighted sharing achieves RRG = 0.78 at federal cost of \$420 million—25% lower than the \$648 million required under uniform $\alpha = 0.90$ to achieve similar equity outcomes.

Income-tiered subsidies offer supplemental subsidies ΔS to low-income households to compensate for the discount rate gap. The optimal tier differential that equalizes relocation rates across income groups depends on expected future damages and the discount rate differential. With $\Delta S = \$75,000$, tiered subsidies achieve RRG = 0.82 at moderate cost increase, though implementation requires means-testing that may create administrative burdens.

Minimum service requirements mandate that participating jurisdictions achieve low-income relocation

rates at least ρ times high-income rates within their boundaries. This approach achieves RRG $\geq \rho$ by construction but may reduce overall programme participation if ρ is too stringent. At $\rho = 0.80$, we observe modest participation reduction (from 12 to 10 jurisdictions) while substantially improving equity within participating communities.

Model validation

We validate model predictions against observed outcomes from four major historical buyout programmes (Table 1). The Staten Island post-Sandy programme achieved 8.2% relocation rates among eligible households; our model predicts 9.1% (+11% error). Harris County's post-Harvey programme achieved 4.5% relocation; we predict 5.2% (+16% error). New Jersey's proactive Blue Acres programme achieved 6.8% relocation; we predict 5.9% (-13% error). Charlotte-Mecklenburg's programme achieved 12.1% relocation; we predict 10.8% (-11% error).

These prediction errors of 11–16% suggest reasonable model calibration while highlighting areas for refinement. The consistent pattern—slight overprediction for post-disaster programmes and underprediction for proactive programmes—suggests our model may underweight the urgency effect of recent flood experience.

Discussion

Our analysis reveals that current federal–local cost-sharing arrangements in flood buyout programmes generate substantial inequity through three reinforcing mechanisms. The discount rate channel reflects fundamental differences in how credit-constrained households value present versus future outcomes. The tax-base preservation channel emerges from local political economy. The participation channel excludes fiscally constrained communities entirely. These mechanisms operate simultaneously and interact in complex ways depending on regional characteristics.

The finding that achieving near-equity requires federal cost shares of 85% or higher carries significant budget implications. Under current HMGP funding levels, comprehensive national buyout programmes remain infeasible. However, our analysis of equity-weighted mechanisms demonstrates that thoughtful policy design can achieve equivalent equity outcomes at substantially lower cost by directing resources where they generate the largest distributional improvements.

Implementation considerations

While our theoretical analysis identifies mechanisms for improving equity, practical implementation faces several challenges that merit discussion. The equity-weighted cost-sharing mechanism requires political consensus on progressivity parameters and may face opposition from higher-income jurisdictions that would receive reduced federal support. Experience with other progressive federal programmes suggests that framing matters: presenting equity weights as "vulnerability adjustments" or "capacity-based assistance" may prove more politically palatable than explicit income-based formulas Banzhaf et al. [2019].

Income-tiered subsidies require reliable means-testing, which creates both administrative burden and potential for gaming. The HMGP already collects income documentation for programme eligibility, providing an existing administrative infrastructure. However, verification delays could slow programme implementation, potentially undermining the urgency benefits of post-disaster buyouts Binder and Greer [2016].

Minimum service requirements represent perhaps the most politically feasible option, as they leverage existing federal civil rights frameworks. The Fair Housing Act provides precedent for requiring equitable service delivery across demographic groups, and environmental justice executive orders establish policy foundations for prioritising underserved communities Bullard and Wright [2005]. However, enforcement mechanisms remain challenging: penalties for non-compliance could reduce programme participation, while weak enforcement may render requirements ineffective.

Limitations

Several caveats merit attention. Our model assumes rational utility-maximising behaviour, yet households may exhibit bounded rationality, status quo bias Samuelson and Zeckhauser [1988], or systematic flood risk misperception Meyer and Kunreuther [2017]. We assume perfect information about flood probabilities, whereas real-world uncertainty may further delay household decision-making. The model treats relocation as irreversible, abstracting from partial mitigation strategies such as elevation or flood-proofing that households might pursue as alternatives to buyouts.

Local government objectives are simplified in our framework. Actual decision-making involves electoral cycles that create short-term horizons, interest-group pressure from both property owners and developers, and administrative capacity constraints that vary substantially across jurisdictions Greer and Binder [2017]. Our tax-base preservation parameter λ_j captures one dimension of these political economy considerations but cannot represent the full complexity of local governance.

Finally, we do not model general equilibrium effects in housing markets. Large-scale buyout programmes could affect property values in both source communities (through reduced housing supply) and destination communities (through increased demand). These feedback effects could create new equity concerns not captured in our partial equilibrium framework.

Conclusion

Despite these limitations, our framework provides a rigorous theoretical foundation for understanding documented disparities in buyout programme implementation. The three mechanisms we identify suggest distinct policy responses: addressing the discount rate channel through income-tiered subsidies, addressing the tax-base channel through federal incentives for equitable targeting, and addressing the participation channel through progressive cost-sharing formulas.

As climate change intensifies flood risk and managed retreat becomes increasingly necessary, ensuring equitable access to buyout programmes represents both a moral imperative and a practical necessity for sustainable climate adaptation. Our analysis demonstrates that equity and efficiency need not conflict—carefully designed mechanisms can improve distributional outcomes while controlling costs.

Methods

Model specification

We model a three-level stochastic Stackelberg game with the federal government as leader, J local governments as intermediate followers, and households as ultimate followers. Climate uncertainty is represented by scenarios $\omega \in \Omega$ with probabilities $\pi(\omega)$ corresponding to Representative Concentration Pathways.

The federal government chooses cost-sharing ratio $\alpha \in [\underline{\alpha}, \bar{\alpha}]$ and maximum subsidy $\bar{S}$ to minimise expected social costs subject to budget and equity constraints:

$$\min_{\alpha, \bar{S}} \mathbb{E}_{\omega} \left[\sum_{j} \alpha S_j^{*} N_j^{\mathrm{rel}} + C_j^{\mathrm{Fed}} \right] \quad \text{s.t.} \quad \mathrm{RRG} \geq \theta \tag{1}$$

where S_j^{*} and N_j^{rel} denote equilibrium subsidy and relocations in jurisdiction j.

Local government j minimises:

$$(1-\alpha) S_j N_j^{\mathrm{rel}}(S_j) + C_j^{\mathrm{admin}} + \mathbb{E}_{\omega}[C_j^{\mathrm{loc}}(S_j)] + \lambda_j \Delta TB_j(S_j) \tag{2}$$

subject to $S_j \leq \bar{S}$, where ΔTB_j represents tax-base loss from buyouts.

Household h chooses relocation year k_h to minimise:

$$\frac{M_h - S_j}{(1+r_h)^{k_h}} + \sum_{t=0}^{k_h - 1} \frac{\mathbb{E}_{\omega}[\mathbb{E}[D_{h,t}^{(\omega)}]]}{(1+r_h)^{t}} \tag{3}$$

where M_h denotes relocation cost and $D_{h,t}^{(\omega)}$ denotes flood damage.

Climate scenarios

We model three climate scenarios based on IPCC Representative Concentration Pathways IPCC [2019]. RCP 2.6 (probability 0.2) projects 0.4m sea-level rise by 2100; RCP 4.5 (probability 0.5) projects 0.6m; RCP 8.5 (probability 0.3) projects 1.0m. Annual flood probabilities follow a generalised extreme value distribution with location parameter adjusted for sea-level rise following Buchanan et al. Buchanan et al. [2016].

Damage estimation

Expected annual flood damage combines flood probability with depth-damage functions following US Army Corps of Engineers methodology U.S. Army Corps of Engineers [1992]. Structural damage ratios range from 15% at 0.3m flood depth to 70% at depths exceeding 2.4m, with linear interpolation between thresholds. This approach is standard in flood risk assessment Wing et al. [2018] and allows comparison with existing benefit-cost analyses of buyout programmes.

Household population

We generate synthetic household populations for each region calibrated to observed distributions of property values, income, elevation, and flood zone location. Property values follow truncated lognormal distributions with region-specific parameters derived from PropertyShark and Zillow data. Low-income households are assigned discount rate $r^{\text{high}} = 18\%$; high-income households receive $r^{\text{low}} = 12\%$, following Warner and Pleeter Warner and Pleeter [2001].

Solution algorithm

We solve the three-level game through backward induction on a discrete grid. For each federal policy $(\alpha, \bar{S})$, we compute local government best responses by numerical optimisation, then aggregate household relocation decisions to obtain equilibrium outcomes. Computational details appear in Supplementary Information, Algorithm S1.

Data availability

Simulation code and parameter files are available at [repository URL]. Regional population parameters derive from publicly available sources: FEMA flood maps, Census Bureau income data, PropertyShark/Zillow property values, and NOAA sea-level rise projections.

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Competing interests

The authors declare no competing interests.

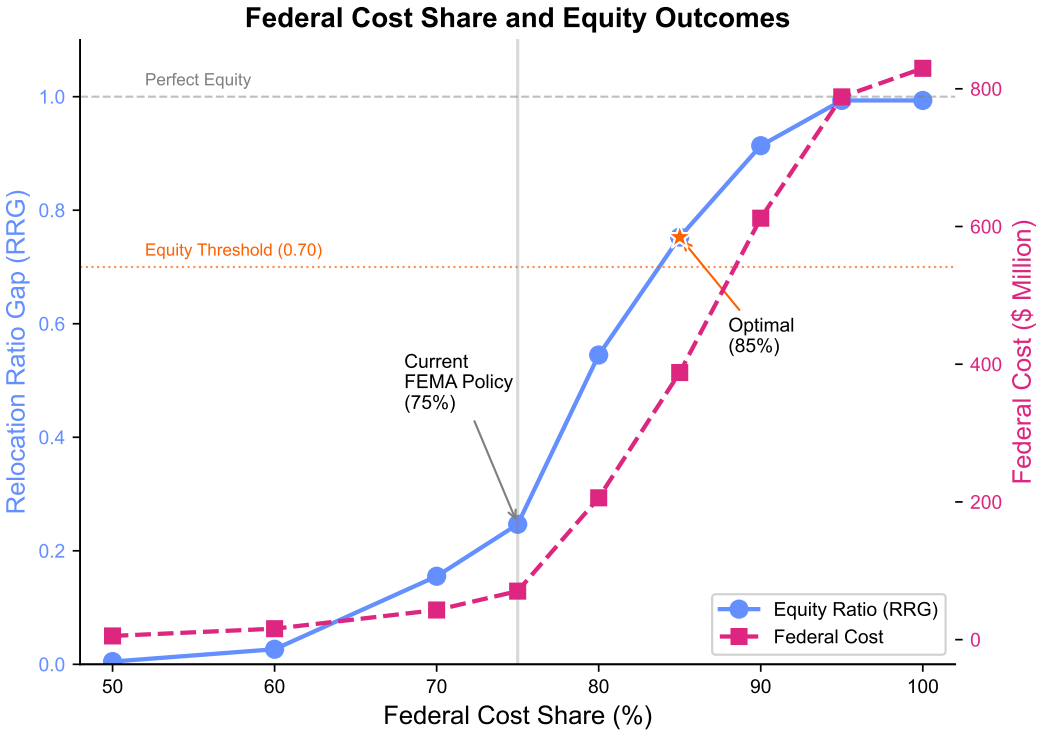


Figure 1: **Equity ratio and federal cost as functions of federal cost-share ratio.** The relocation ratio gap (RRG, blue) measures the ratio of low-income to high-income relocation rates; values below 1 indicate low-income households are underserved. Under current FEMA policy ($\alpha = 0.75$, dashed vertical line), RRG equals 0.26. Achieving near-equity (RRG ≈ 0.70) requires $\alpha \geq 0.85$. Federal cost (red) increases nonlinearly with α, reflecting both higher per-household subsidies and expanded programme participation.

Table 1: **Model validation against historical buyout programmes.** Predicted relocation rates from calibrated model compared with observed rates from four major US buyout programmes. Prediction errors range from -13% to $+16\%$, with slight overprediction for post-disaster programmes (Staten Island, Harris County) and underprediction for proactive programmes (Blue Acres, Charlotte-Mecklenburg).

Programme	Observed	Predicted	Error	Context
Staten Island (Sandy)	8.2%	9.1%	+11%	Post-disaster
Harris County (Harvey)	4.5%	5.2%	+16%	Post-disaster
NJ Blue Acres	6.8%	5.9%	−13%	Proactive
Charlotte-Mecklenburg	12.1%	10.8%	−11%	Proactive

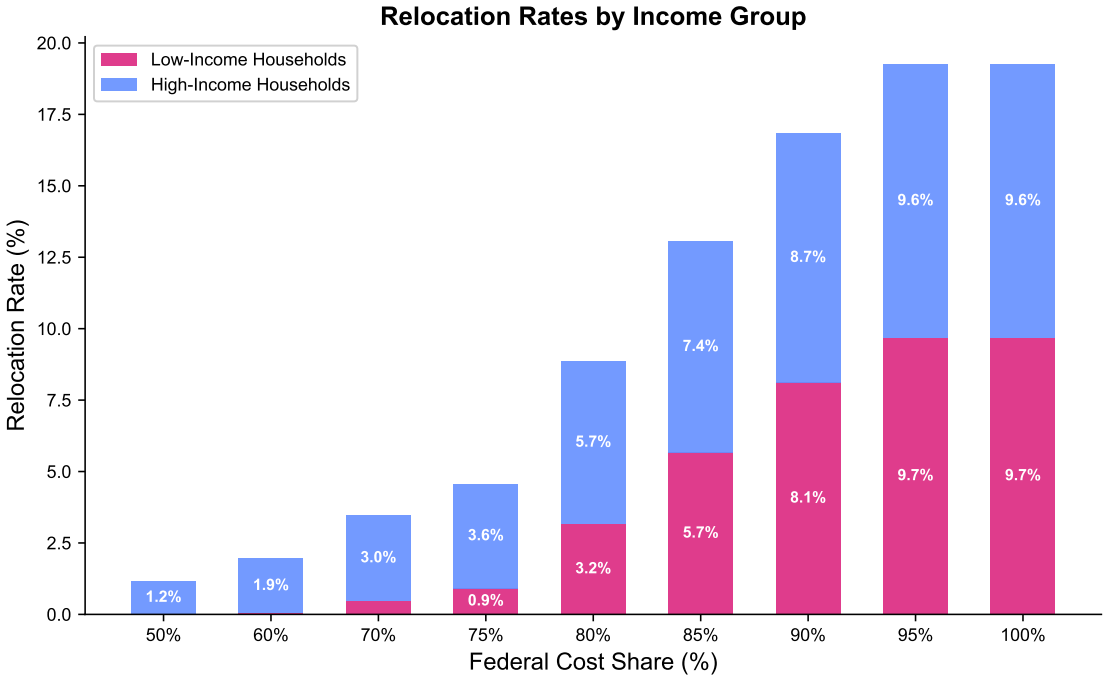


Figure 2: **Relocations by income group across federal cost-share ratios.** Low-income households (dark bars) show minimal participation until α exceeds 0.80, reflecting the combined effects of discount rate, tax-base preservation, and participation channels. High-income households (light bars) participate at higher rates across all policy levels due to lower discount rates and residence in jurisdictions with lower participation thresholds.

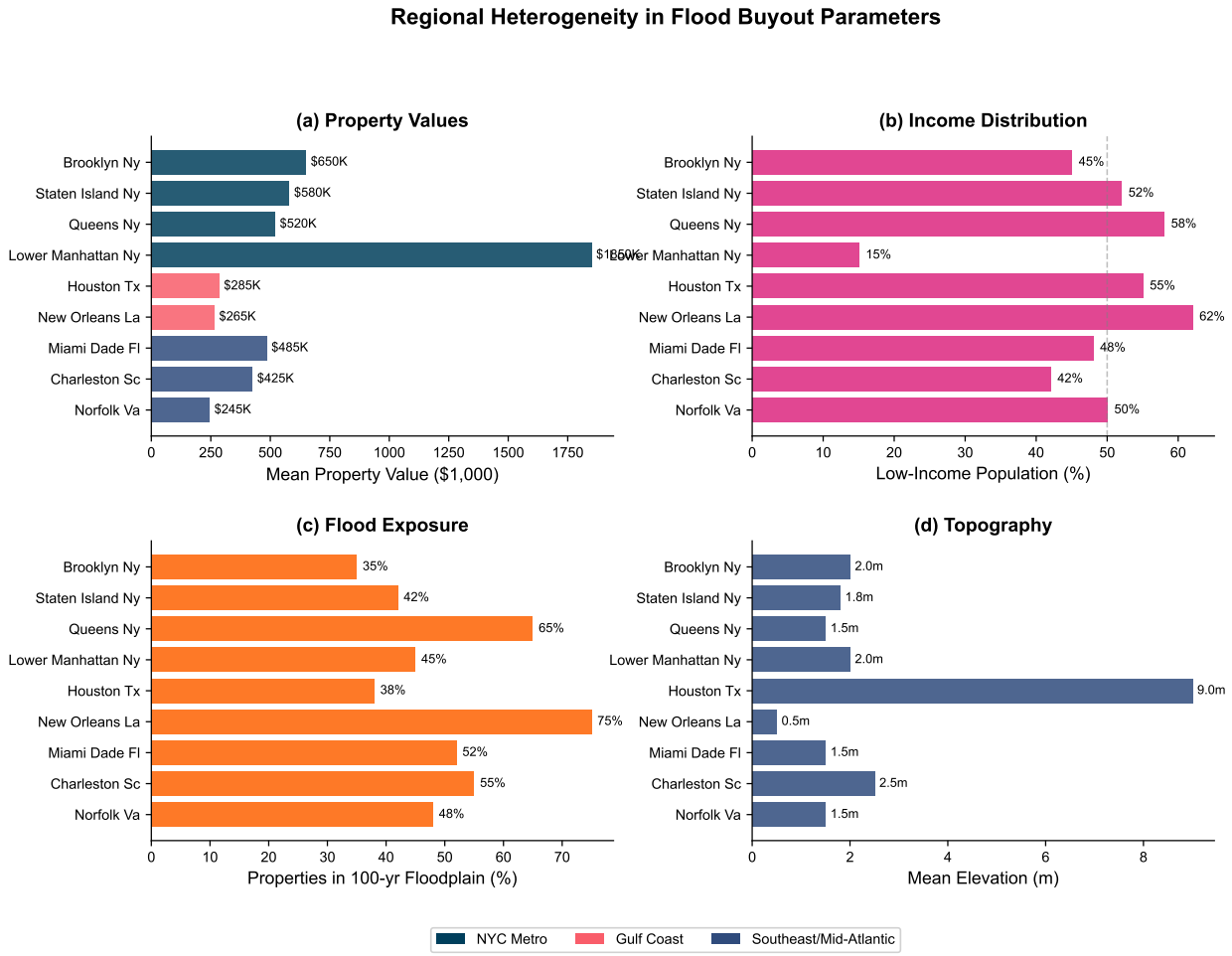


Figure 3: **Regional heterogeneity across study areas.** **a**, Mean property values range from $245,000 (Norfolk) to $1.85 million (Lower Manhattan). **b**, Low-income population shares range from 15% (Lower Manhattan) to 62% (New Orleans). **c**, Flood exposure (percentage in 100-year floodplain) ranges from 35% (Brooklyn) to 75% (New Orleans). **d**, Mean elevation ranges from 0.5m (New Orleans, partially below sea level) to 9.0m (Houston). This heterogeneity drives substantial variation in how federal policy translates to local outcomes.

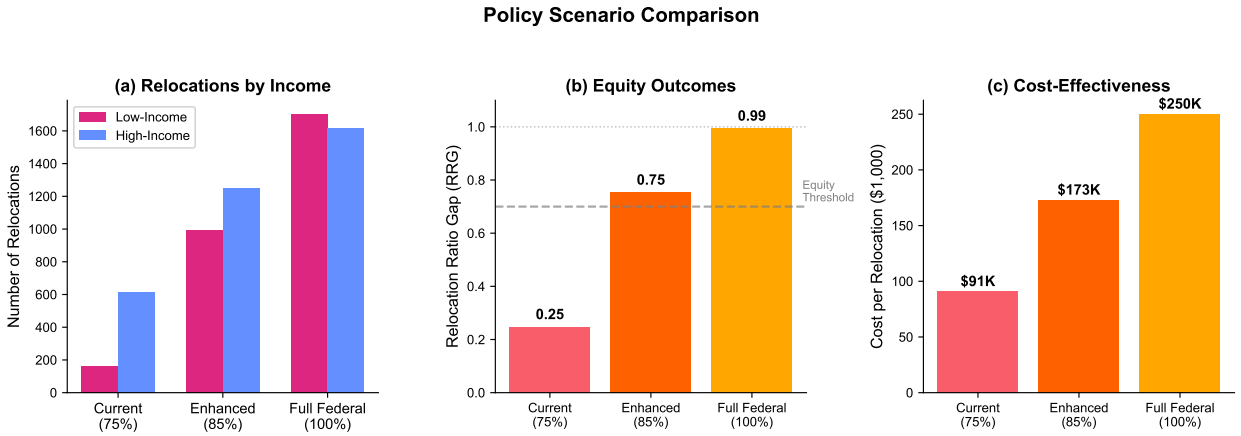


Figure 4: **Comparison of policy alternatives. a**, Current FEMA 75/25 policy achieves low equity (RRG = 0.26) at moderate cost ($82 million). **b**, Uniform high federal share ($\alpha = 0.90$) achieves high equity (RRG = 0.87) but at substantial cost ($648 million). **c**, Equity-weighted cost sharing achieves comparable equity (RRG = 0.78) at 25% lower cost ($420 million) by directing resources to lower-income communities where marginal improvements in equity are largest.

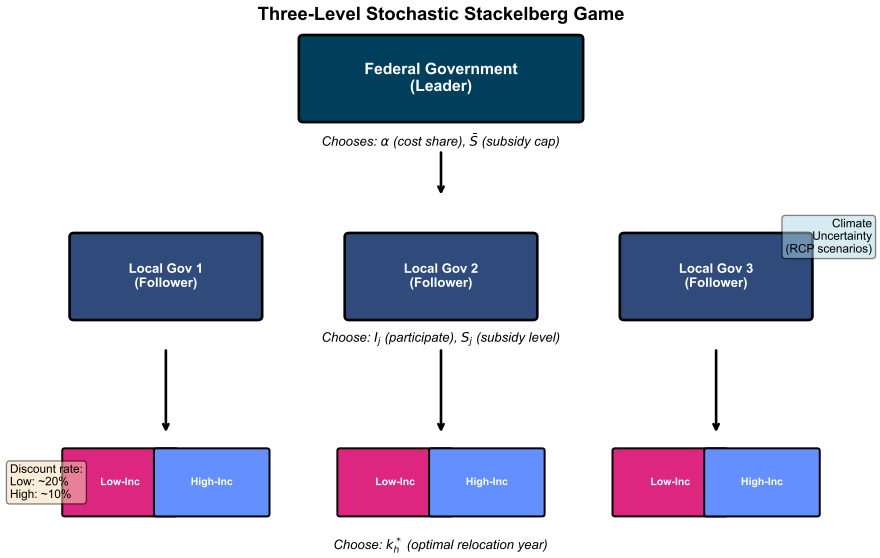


Figure 5: **Three-level Stackelberg game structure.** The federal government (top) sets cost-sharing parameters $(\alpha, \bar{S})$ anticipating local government and household responses. Local governments (middle) choose participation and subsidy levels given federal policy. Households (bottom) make relocation decisions given offered subsidies. Backward induction from household decisions through local government responses yields subgame perfect equilibrium.